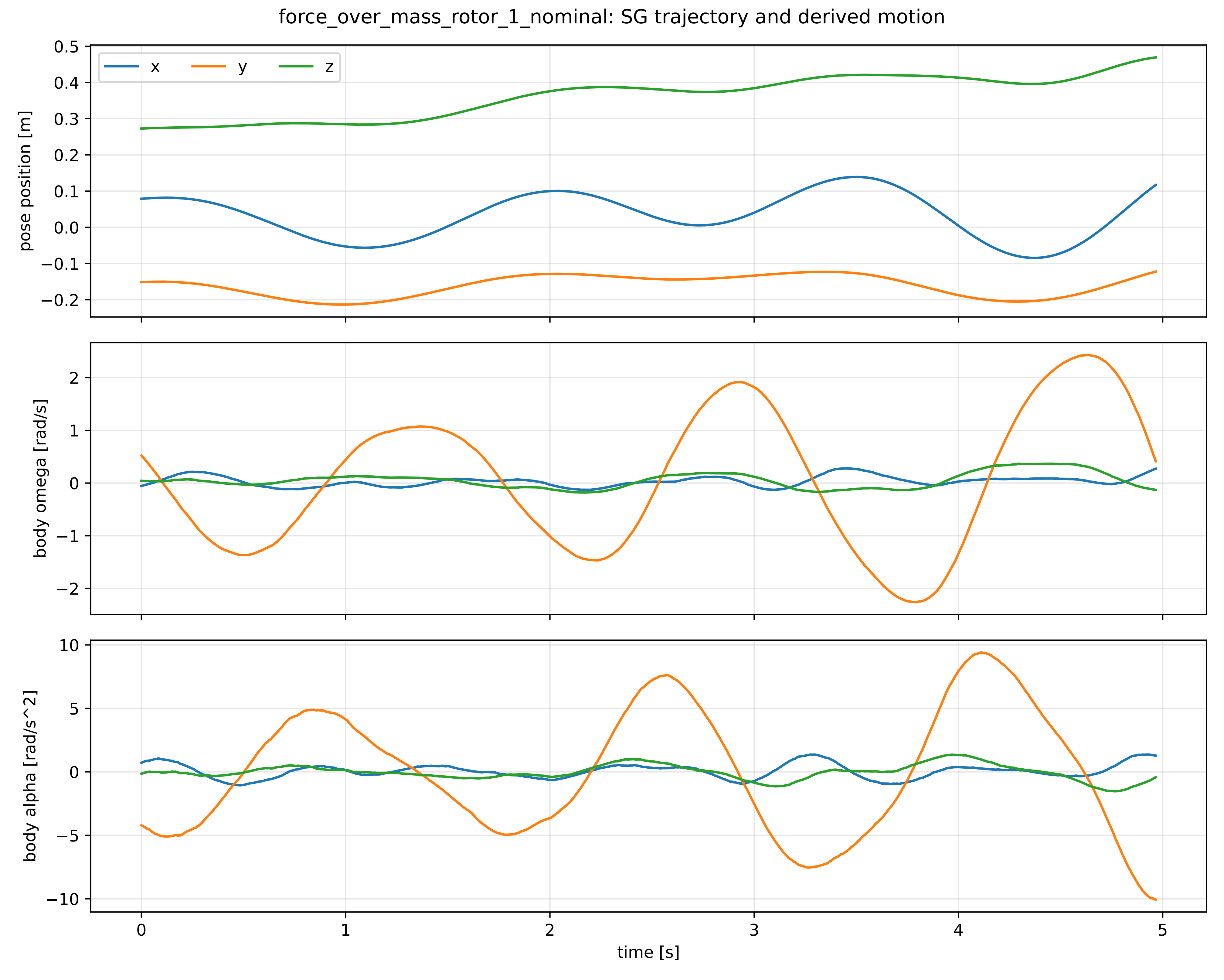
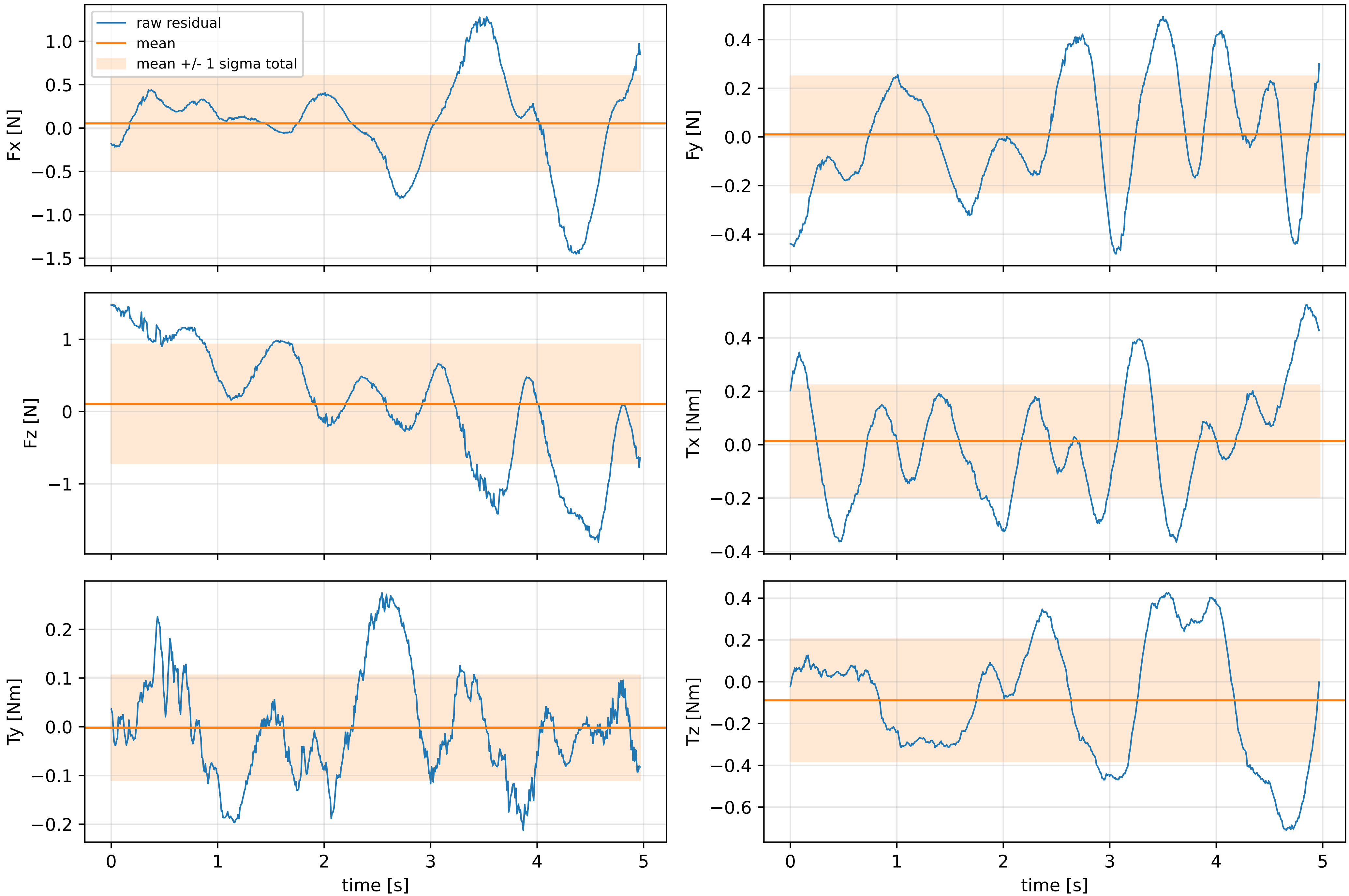
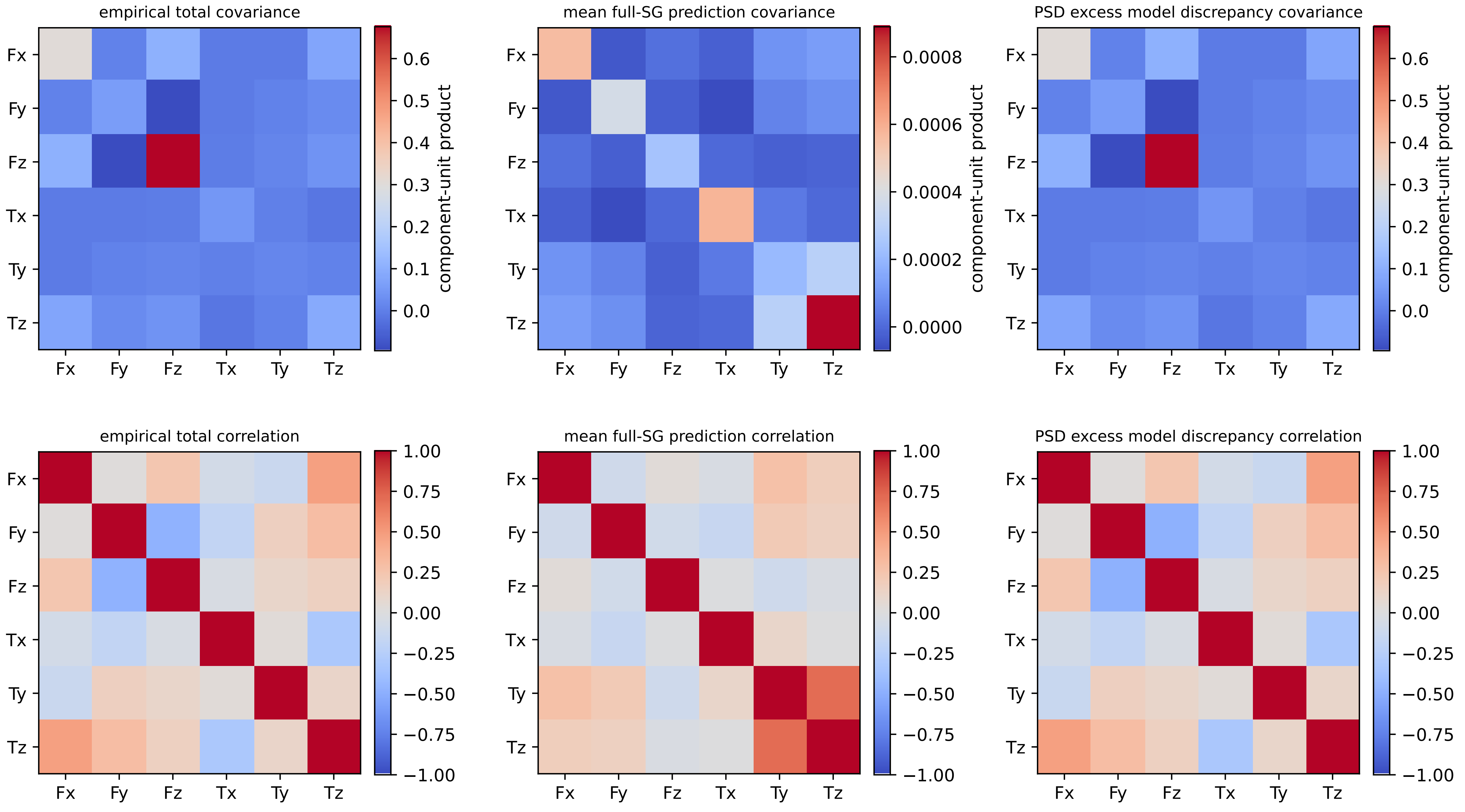

force_over_mass_rotor_1_nominal: SG trajectory and derived motion
x
y
z
pose position [m]
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
body omega [rad/s]
2
1
0
−1
−2
body alpha [rad/s^2]
10
5
0
−5
−10
0
1
2
3
4
5
time [s]

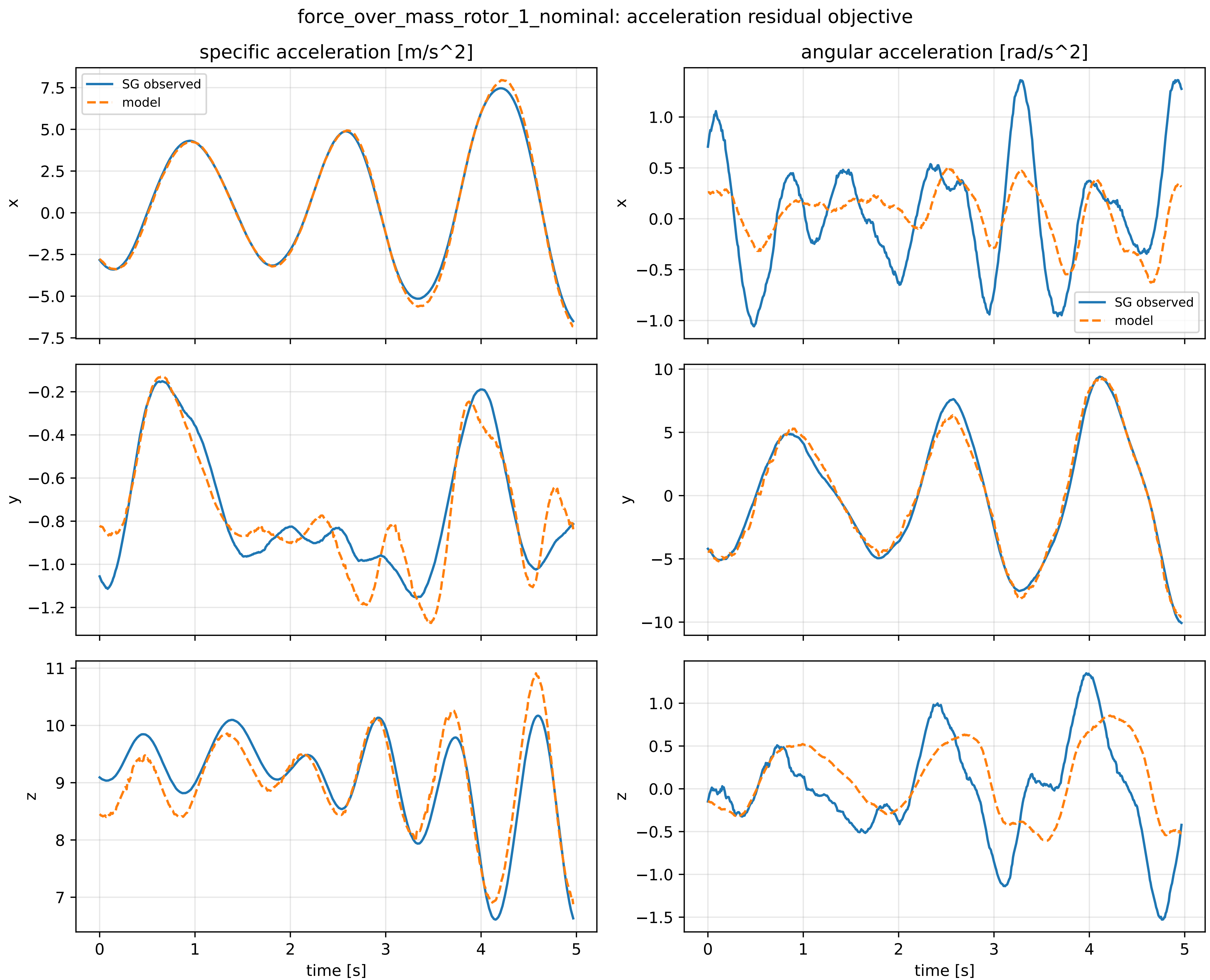
force_over_mass_rotor_1_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg

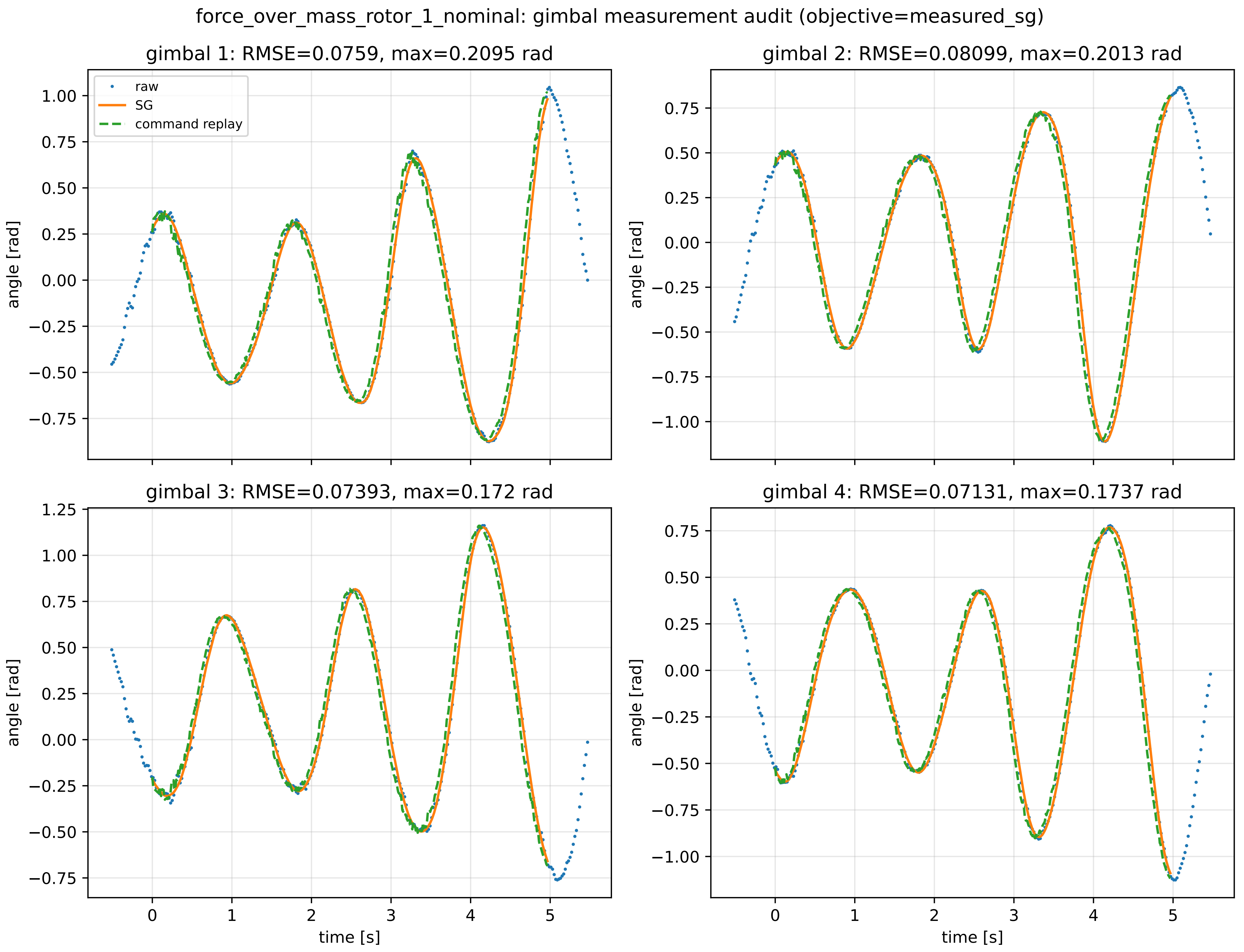


vector RMS about zero: force 1.0228 N, torque 0.3855 Nm; component std about mean: force [0.5483 0.239 0.8221], torque [0.2088 0.1075 0.2926]

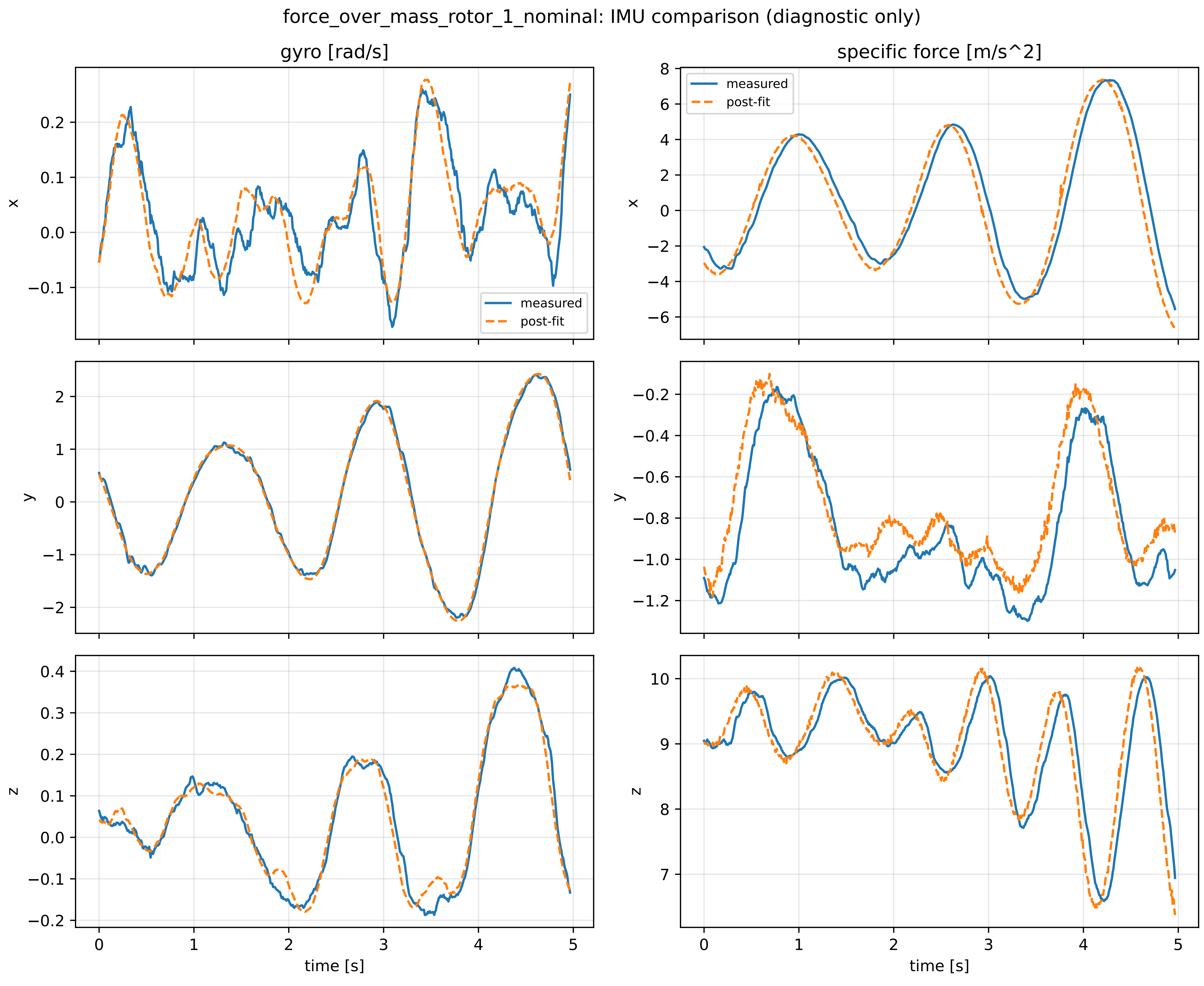
force_over_mass_rotor_1_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.01 0.029 0.034 0.082 0.298 0.719]



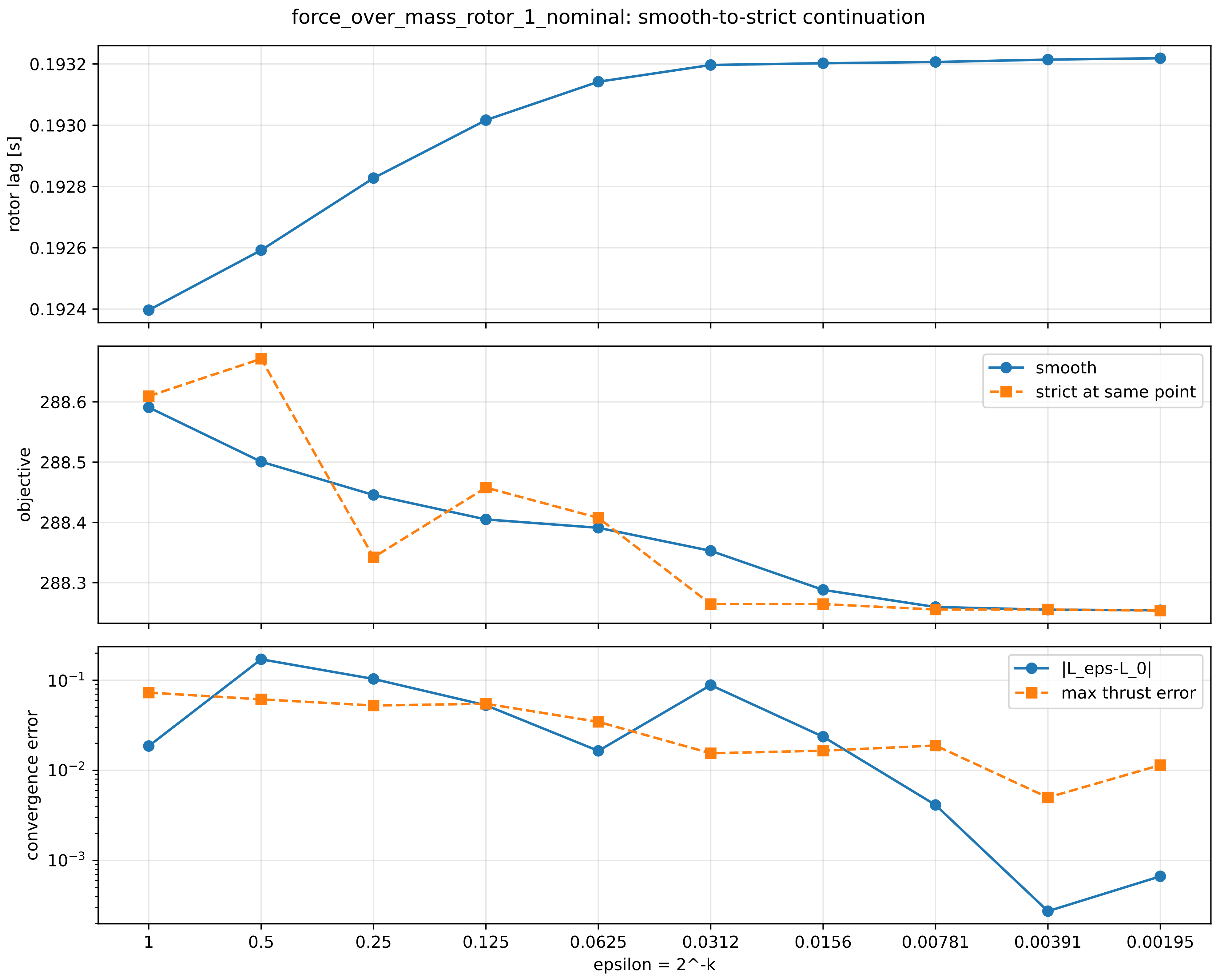

force_over_mass_rotor_1_nominal: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]



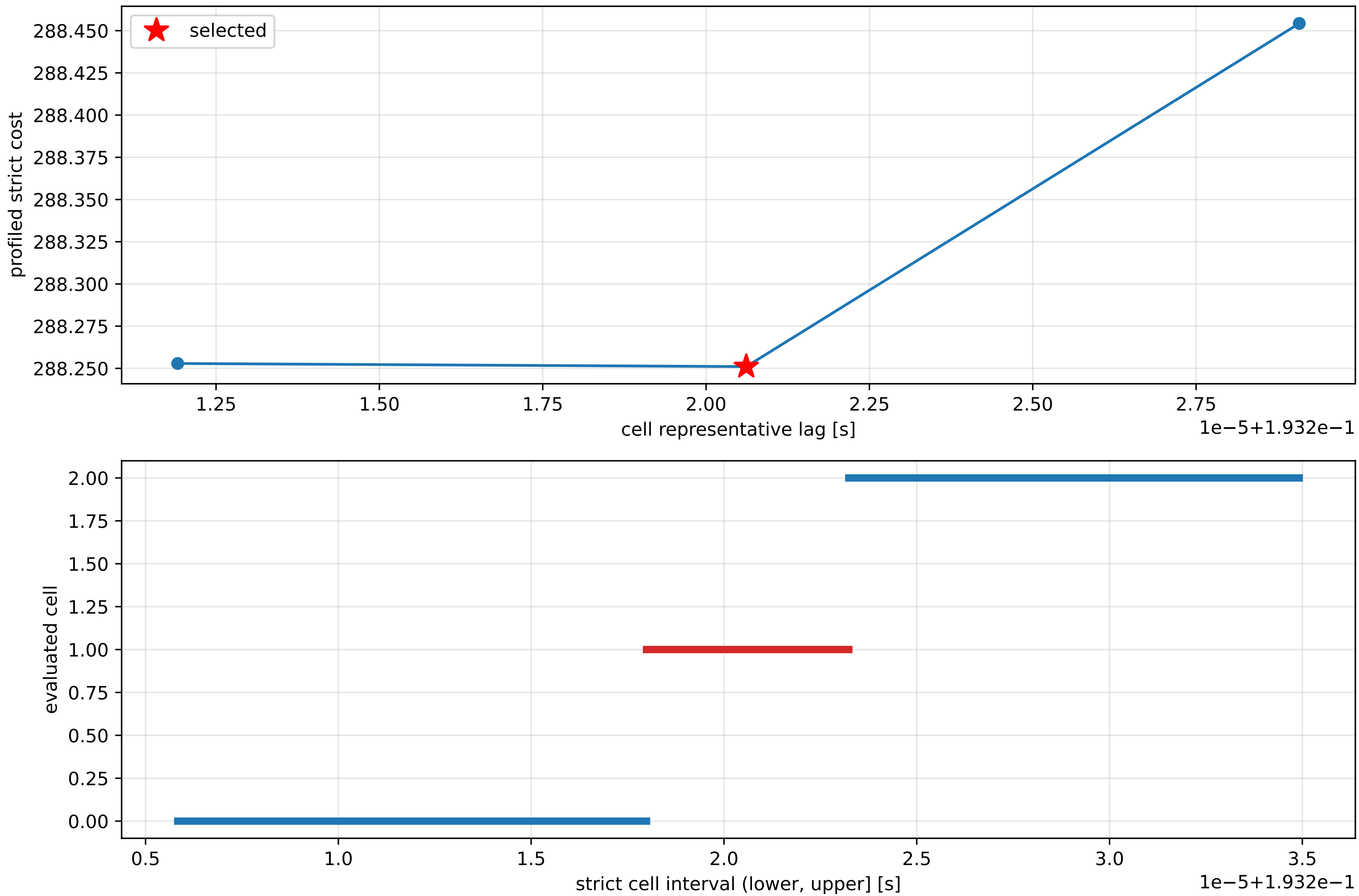
force_over_mass_rotor_1_nominal: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.0759, max=0.2095 rad
gimbal 2: RMSE=0.08099, max=0.2013 rad
gimbal 3: RMSE=0.07393, max=0.172 rad
gimbal 4: RMSE=0.07131, max=0.1737 rad
raw
SG
command replay
angle [rad]
time [s]



force_over_mass_rotor_1_nominal: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]


force_over_mass_rotor_1_nominal: smooth-to-strict continuation
rotor lag [s]
0.1932
0.1930
0.1928
0.1926
0.1924
objective
288.6
288.5
288.4
288.3
smooth
strict at same point
convergence error
10^-1
10^-2
10^-3
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k

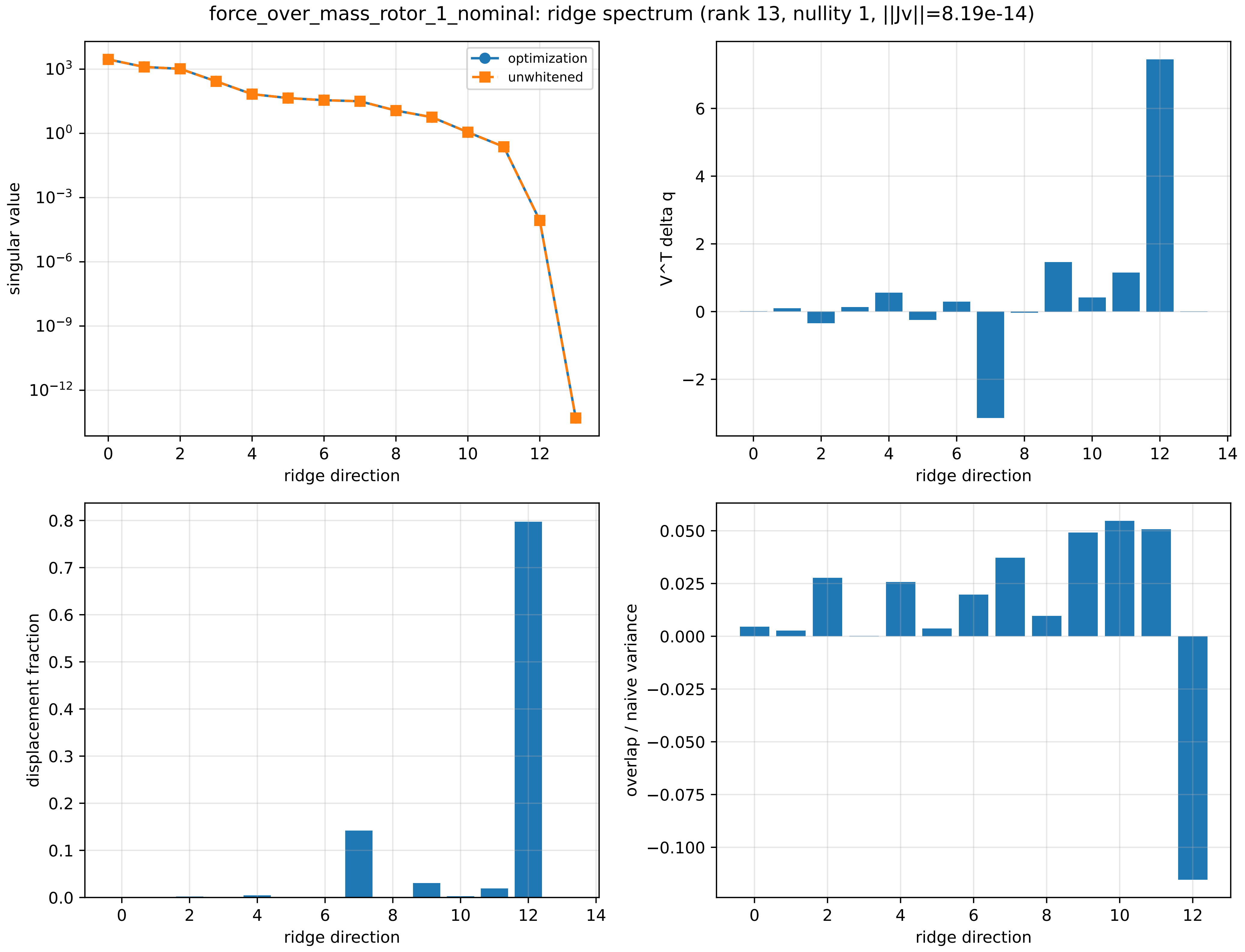
force_over_mass_rotor_1_nominal: exact strict-ZOH lag cells

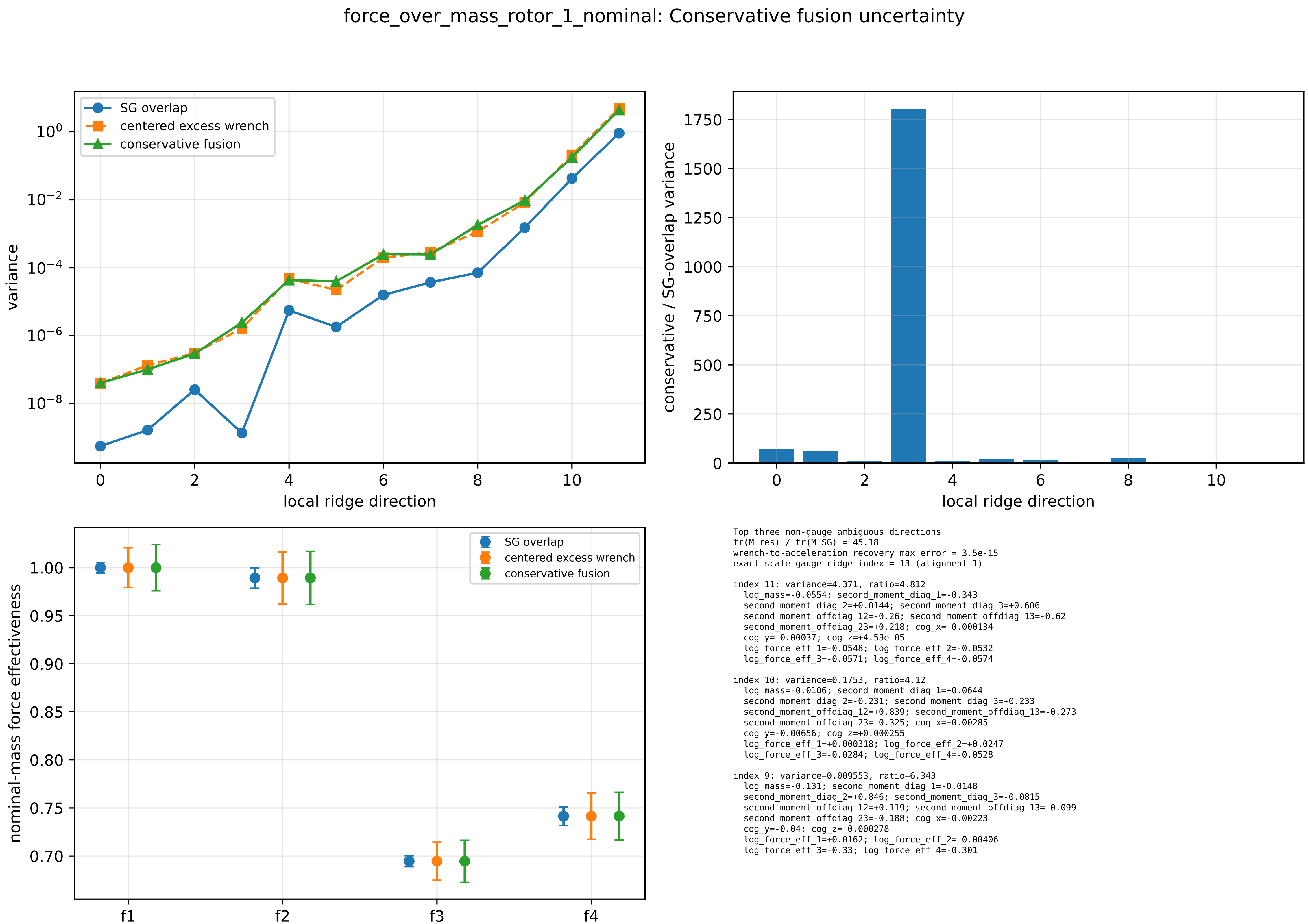


selected (0.193217993, 0.193223238] s; final smooth 0.193218651 s; data boundary=False

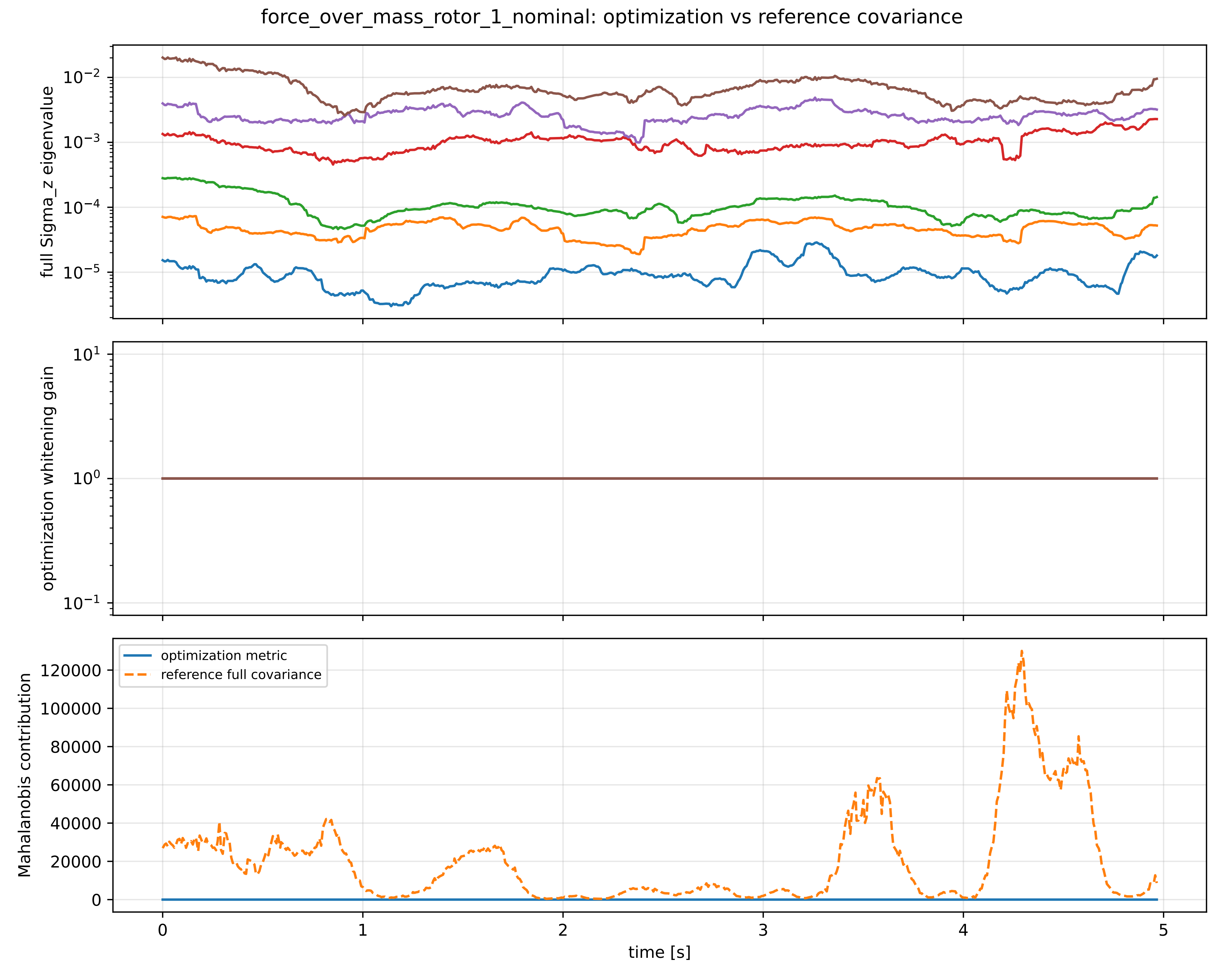
force_over_mass_rotor_1_nominal: parameter estimate

```
Case: force_over_mass_rotor_1_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.15261536
inertia [kg m^2]:
[[ 6.04227497e-01  4.07775201e-04 -7.33398379e-02]
 [ 4.07775201e-04  2.53235163e-01  2.57954837e-02]
 [-7.33398379e-02  2.57954837e-02  8.08109374e-01]]
CoG body [m]: [-0.01104793 -0.03070114 -0.0494818 ]
force effectiveness: [1.34062697 1.32629329 0.9311412  0.99398511]

scale-free J/m [m^2]:
[[ 1.91659123e-01  1.29345053e-04 -2.32631734e-02]
 [ 1.29345053e-04  8.03254232e-02  8.18224894e-03]
 [-2.32631734e-02  8.18224894e-03  2.56329834e-01]]
scale-free f/m: [0.4252428  0.4206962  0.29535515 0.31528905]

rotor lag cell: (0.193217993, 0.193223238] s
representative: 0.193220615 s
```

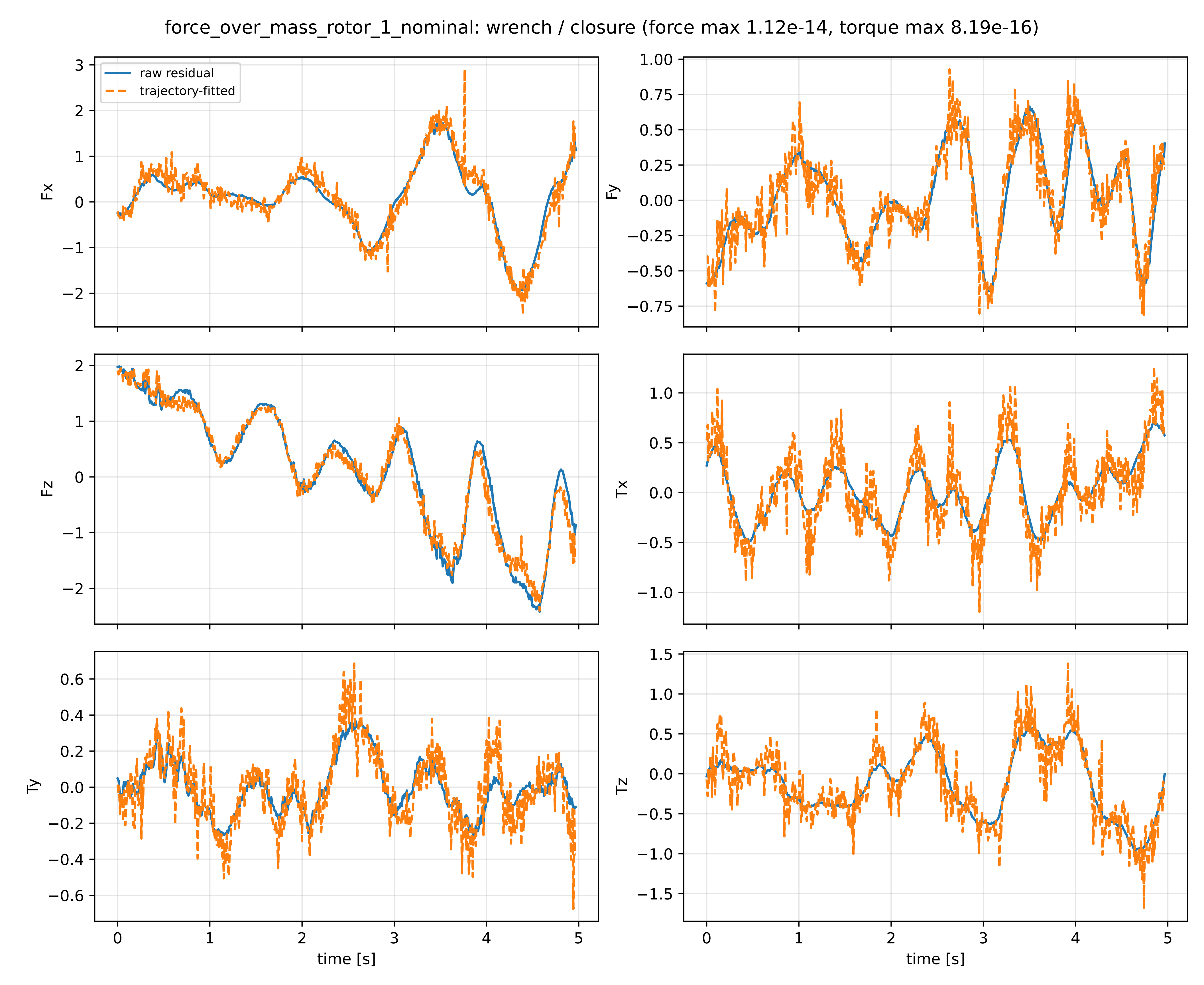

force_over_mass_rotor_1_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=8.19e-14)
optimization
unwhitened
singular value
ridge direction
V^T delta q
ridge direction
displacement fraction
ridge direction
overlap / naive variance
ridge direction



The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.



force_over_mass_rotor_1_nominal: optimization vs reference covariance
full Sigma_z eigenvalue
optimization whitening gain
Mahalanobis contribution
optimization metric
reference full covariance
time [s]



force_over_mass_rotor_1_nominal: wrench / closure (force max 1.12e-14, torque max 8.19e-16)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]

force_over_mass_rotor_1_nominal: optional parameter-prior audit

```
Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.
name: pseudo_condition_force_over_mass_rotor_1_nominal
role: pseudo_conditioning_ablation
source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/force_over_mass
SHA256: 6d6d7476c7c2cc90b6dbb115ea73393bd5979488f12a4de1ccf810600f0d2db0

data objective:  288.251046797
prior objective: 3.94423443405e-06
total objective: 288.251050741

This configuration uses an intentionally tight artificial standard deviation
for an ablation-style pseudo-conditioning experiment; it is not a calibrated
physical uncertainty.

factor: force_over_mass_rotor_1_nominal (force_effectiveness_over_mass)
  components: rotor_1
  target: [0.42524282]
  std:    [1.e-05]
  final:  [0.4252428]
  error:  [-2.80864182e-08]
  standardized residual: [-0.00280864]
  factor objective: 3.94423443405e-06
```